EXPERIMENT-16 Feed Forward Neural Network

Aim

To implement a **Feed Forward Neural Network (FFNN)** using Python.

Objectives

- To understand the architecture of neural networks.
- To implement input, hidden and output layers.
- To train the network using sample data.
- To perform prediction.

Algorithm

1. Import NumPy and neural network libraries.
2. Create input and target datasets.
3. Define the neural network.
4. Add input, hidden and output layers.
5. Train the network.
6. Give test input.
7. Predict the output.
8. Display the result.

Flowchart

```
  START
    |
    v
Input data
    |
    v
Input Layer
```

|

v

Hidden Layer

|

v

Output Layer

|

v

Calculate Error

|

v

Train Network

|

v

Prediction

|

v

STOP

Python Program

```
import numpy as np
from sklearn.neural_network import MLPClassifier


# XOR training data
X = np.array([
    [0, 0],
```

```
    [0, 1],
    [1, 0],
    [1, 1]
])

y = np.array([0, 1, 1, 0])

# Create feed-forward neural network
model = MLPClassifier(
    hidden_layer_sizes=(4,),
    activation='relu',
    solver='lbfgs',
    max_iter=1000,
    random_state=1
)

# Train the network
model.fit(X, y)

# Test data
test = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
```

```
])
```

```
# Prediction
prediction = model.predict(test)

print("Input:")
print(test)

print("Predicted Output:")
print(prediction)
```

output

Result

The Feed Forward Neural Network successfully learned the XOR pattern and predicted the outputs.

Conclusion

Thus, a Feed Forward Neural Network was successfully implemented and trained for classification.